Zhexiang (Bruce) Li

bruceli@ucla.edu | +1 (831) 239-2728 | Los Angeles, CA | brucelzx.github.io | github.com/BruceLZX | linkedin.com/in/zhexiang-li

Available for internships: Jun 13 – Sep 17, 2027

Education

University of California, Los Angeles Expected Jun 2028
M.S. in Computer Science

University of Southern California May 2026
B.S. in Computer Science, *Magna Cum Laude*; GPA: 3.76/4.00; Viterbi Dean's List

Technical Skills

Languages: Python, C/C++, JavaScript/TypeScript, SQL, Java, Swift
AI/ML: PyTorch, Triton, LLM APIs, agent workflows, RAG (BGE embeddings, ChromaDB), multimodal inference
Applications: FastAPI, asyncio, SQLAlchemy, MySQL, PostgreSQL (Supabase), Next.js, Docker, Linux, Git

Engineering & Research Experience

Health Intelligence Lab, UCLA | *Graduate Researcher* Aug 2026 – Present

- Curated a unified CGM release from **49 datasets** (**7,000+** source person records) with meal, insulin, and wearable context, provenance checksums, and 14 automated release checks.
- Built patient-disjoint benchmarks and data audits with Prof. Yuzhe Yang to compare GluFormer, T1DFormer, and custom glucose forecasters at 30-, 60-, and 120-minute horizons.

PMAOS / Next Big Seek | *AI Engineer Intern* Jul – Aug 2026

- Designed short-video recommendation using bidirectional user–video interest propagation, exposure budgets across multiple timescales, and fatigue controls to manage repeated recommendations.
- Built **11-language** audio-visual affect inference that fuses evidence and abstains when modalities conflict.

Terlingua Threads | *Full-Stack Engineer* Remote · Jan – May 2026

- Built Shopify-integrated pages and Google Maps features with responsive, accessible layouts.

Piezolab Medical | *AI Engineer Intern* Lake Forest, CA · Jun – Aug 2025

- Built an LLM-assisted pipeline to ingest, clean, and cluster **5,000+** medical-malpractice documents for quality-of-care review; developed agent workflows and acceptance tests for vascular-care research.

Selected AI Projects

AI Doctor WeChat | *Python, FastAPI, MySQL, RAG* 2026

- Built and deployed a FastAPI backend (**110** endpoints, **26** MySQL tables) on WeChat Cloud Hosting for a clinic's Mini Program and web workspace covering clinical teaching and patient intake.
- Added five-role access control with doctor sign-off before patients see records, RAG (BGE embeddings, ChromaDB, keyword fallback), and an asyncio queue that rate-limits LLM calls globally, per user, and per API.

Fridge Manager | *Next.js, TypeScript, Supabase, LLM and vision models* 2026

- Built and launched howtoeat.food, an AI kitchen web app that plans meals and grocery lists from each user's inventory, leftovers, preferences, cookware, and cooking time.
- Shipped photo-based inventory recognition and LLM recipe import with user corrections, admin review, and deduplication; a multimodal video pipeline turned **758** cooking videos into **121** recipes.

Investment Analysis Agent Skill | *Python, Claude Code* Jul 2026

- Wrote an agent skill that turns a company name or pitch deck into an evidence-backed report across 14 dimensions; a Python QA harness gates each section on independent and primary sources, depth, and completeness.

Selected ML Research

- **Efficient Attention via Pre-Scoring** (first author; under review): Triton key selection that, with HyperAttention, cut ChatGLM perplexity from **12.0 to 9.5** at 131k-token contexts under a fixed interaction budget. arXiv:2505.11040
- **MedTokenBudget** (sole author; preprint): lesion-supervised post-backbone token routing reached **80.38%** mean accuracy on ISIC 2019 at a 30% token budget versus 76.93% for ToMe across three seeds. arXiv:2609.30613